

PUNJAB GREENHOUSING FEATURES

Sector: Infrastructure & Urban Resilience
Hazard addressed: Flood
Targeted assets: Urban Buildings (all types)



Description: This measure involves retrofitting residential buildings with a package of green housing features comprising a green roof, infiltration trench, and rainwater harvesting barrel. The combined package reduces stormwater runoff, mitigates localized flooding, and improves household water security.

Priority Districts: Gujrat, Mandi Bahauddin, Narowal, Sheikhupura, Nankana Sahib, Lahore.

TECHNICAL SPECIFICATIONS/DIMENSION

Unit of implementation: Per building.

Scale assumptions: All residential buildings in target districts.

Key design parameters: Package per building: green roof + infiltration trench + rainwater harvesting barrel.

Time horizon: 2026–2050.

COST ASSUMPTIONS

Unit cost: No single per-building unit cost given; varies by building size/value.

Capital Expenditure (CAPEX):
For Urban Resilience: USD 3 million per year (2026 – 2035). For Infrastructure: USD 37.2 million per year (2026 – 2040).

Operational Expenditure (OPEX):
For Urban Resilience: USD 1.5 million per year (2036–2050). For Infrastructure: USD 25.2 million per year (2036 – 2050).

IMPACT

Risk reduction level: High Effectiveness – significant reduction.

Time to effectiveness: Medium-term; effectiveness begins once green housing features are installed and operational per household; progressive over 2026–2040.

Expected impact: 15% flood intensity reduction; applicable for 1-in-30-year flood events.

IMPLEMENTATION CONSIDERATIONS

Enabling conditions: Financial incentives or subsidies for household adoption; training for building contractors; regulatory standards for green housing features; public awareness campaigns.

Required institutions: Local Government & Community Development Department; Building Control Authorities; Punjab Environment Protection Department.

Barriers: High upfront cost per household without subsidies; low awareness and willingness to adopt; lack of trained contractors; building code enforcement gaps; maintenance neglect reducing long-term effectiveness.

CO-BENEFITS & TRADE-OFFS

Environmental: Reduced stormwater runoff and urban heat island effect (green roofs); improved groundwater recharge (infiltration trenches); reduced pressure on urban drainage systems; potential biodiversity co-benefits from green roofs.

Social: Reduced flood inundation of residential areas; improved household water security through rainwater harvesting; increased community awareness of flood adaptation.

Economic: Reduced flood damage to residential properties; water bill savings from rainwater harvesting; energy savings due to insulation through green roofs; potential increase in property values; reduced burden on municipal drainage infrastructure.

Potential risks: High aggregate CAPEX; requires household-level adoption and maintenance; risk of poor installation quality without training; effectiveness dependent on rainfall patterns and building typology.

BENEFICIARIES

Primary beneficiaries: Residential building owners and occupants in target districts; municipal authorities/utilities benefiting from reduced drainage burden.

Distributional aspects: Benefits primarily accrue to residential building owners, however, all residents benefit from reduced urban flooding and lowered heat island effects.